

**Operations Research
2017/2018**

Exercises Sheet 2

Solving LP problems using the Simplex Method

- 1.** Consider the following linear programming problem:

Maximize $z = 20x_1 + 10x_2$

Subject to

$$x_1 - x_2 \leq 1$$

$$3x_1 + x_2 \leq 7$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic.

- 2.** Consider the following linear programming problem:

Maximize $z = 3x_1 - 2x_2$

Subject to

$$x_1 \leq 4$$

$$x_1 + 3x_2 \leq 15$$

$$2x_1 + x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic.

- 3.** Solve the following linear programming problem by the Simplex method:

Maximize $z = x_1 + 9x_2 + x_3$

Subject to

$$x_1 + 2x_2 + 3x_3 \leq 9$$

$$3x_1 + 2x_2 + 2x_3 \leq 15$$

$$x_1, x_2, x_3 \geq 0$$

- 4.** Consider the following linear programming problem:

Maximize $z = x_1 + 2x_2$

Subject to

$$x_1 + 5x_2 \leq 10$$

$$x_1 + 3x_2 \leq 6$$

$$2x_1 + 2x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic.

- 5.** Consider the following linear programming problem (corresponds to the problem 1-e) of Exercises Sheet 1):

Maximize $z = 2000x_1 + 8000x_2$

Subject to

$$200x_1 + 1600x_2 \leq 4800$$

$$4x_1 + 8x_2 \leq 48$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic.

- 6.** Consider the linear programming problem (corresponds to the problem 1-f of Exercises Sheet 1)

Maximize $z = 4000x_1 + 2500x_2$

Subject to

$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$2x_1 + x_2 \leq 400$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic.

- 7.** Consider the following linear programming problem:

Maximize $Z = 4500x_1 + 4500x_2$

Subject to

$$x_1 \leq 1$$

$$x_2 \leq 1$$

$$5000x_1 + 4000x_2 \leq 6000$$

$$400x_1 + 500x_2 \leq 600$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method, and show the correspondence between the feasible basic solutions and the points of the graphic

8. Consider the following linear programming problem:

Maximize $Z = x_1 + x_2$

Subject to

$$x_1 + 2x_2 \leq 4$$

$$x_1 + x_2 = 3$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method and the “Big M” technique.

9. Consider the following linear programming problem:

Minimize $Z = 3x_1 + 2x_2$

Subject to

$$2x_1 + x_2 \geq 10$$

$$-3x_1 + 2x_2 \leq 6$$

$$x_1 + x_2 \geq 6$$

$$x_1, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the Simplex method and the “Two Phases” technique.

10. Consider the following linear programming problem:

Minimize $Z = 3x_1 + 2x_2 + 4x_3$

Subject to

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 60 \\ 3x_1 + 3x_2 + 5x_3 &\geq 120 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

- Solve the problem using the Simplex method and the “Big M” technique.
- Solve the problem again using the Simplex method but, this time, with the “Two Phases” technique. Compare the sequence of feasible basic solutions obtained in a) and b).

- 11.** Solve the following linear programming problem by the Simplex method using the “Big M” technique:

$$\begin{aligned} \text{Maximize } Z &= 3x_1 + 5x_2 + x_3 \\ \text{Subject to} \\ 2x_1 + 2x_2 + x_3 &\leq 10 \\ -3x_1 + x_2 - x_3 &\geq 6 \\ x_1 \geq 0, x_2 \geq 0, x_3 &\geq 0 \end{aligned}$$

- 12.** Consider the following linear programming problem:

$$\begin{aligned} \text{Minimize } Z &= x_1 - 2x_2 \\ \text{Subject to} \\ 3x_1 - 2x_2 &\leq 9 \\ x_1 + x_2 &\geq 4 \\ -x_1 + 2x_2 &\geq 2 \\ x_1 \geq 0, x_2 &\geq 0 \end{aligned}$$

- Solve it using the graphical method;
- Solve it again using the Simplex method and the “Two Phases” technique.